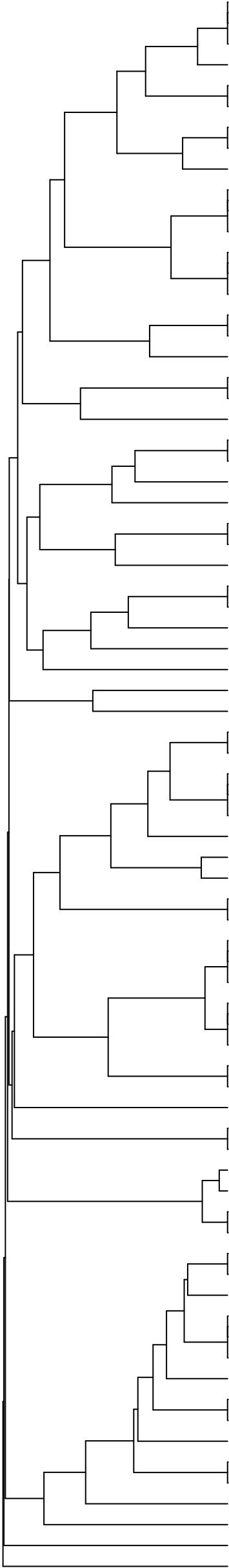


- 17/146 nucleoside triphosphate biosynthetic process
14/108 purine nucleoside triphosphate biosynthetic process
11/24 protein trimerization
13/190 external encapsulating structure organization
7/98 cell wall organization
46/464 meiotic cell cycle
65/589 anatomical structure formation involved in morphogenesis
16/190 sporulation
5/31 response to glucagon
10/73 culmination involved in sorocarp development
18/235 obsolete cellular carbohydrate metabolic process
1/14 positive regulation of DNA endoreduplication
4/29 regulation of DNA endoreduplication
61/538 cellular anatomical entity morphogenesis
16/172 cellular component assembly involved in morphogenesis
59/574 chemical homeostasis
1/21 response to muscle
11/102 positive regulation of transmembrane transport
1/16 regulation of skeletal muscle tissue regeneration
10/86 positive regulation of monotonic ion transmembrane transport
4/47 positive regulation of NF-kappaB transcription factor
4/36 central nervous system neuron differentiation
5/20 cAMP metabolic process
8/27 cyclic nucleotide metabolic process
7/12 cGMP metabolic process
3/18 obsolete cGMP-mediated signaling
11/65 obsolete cyclic-nucleotide-mediated signaling
25/148 obsolete second-messenger-mediated signaling
11/75 calcium-mediated signaling
18/117 response to purine-containing compound
0/24 synaptic transmission, glutamatergic
17/181 locomotory behavior
5/72 neuromuscular junction development
13/57 regulation of striated muscle contraction
18/105 regulation of muscle system process
29/217 regulation of system process
13/90 muscle system process
63/585 system process
50/381 obsolete inorganic ion transmembrane transport
13/62 membrane depolarization
13/89 action potential
68/590 monoatomic ion transmembrane transport
19/108 sodium ion transport
14/120 potassium ion transport
59/485 monoatomic cation transport
41/328 metal ion transport
17/143 calcium ion transport
0/28 ammonium transmembrane transport
9/109 organic cation transport
16/162 lipid localization
15/118 lipid transport
5/10 cholesterol efflux
11/52 organic hydroxy compound transport
3/41 phospholipid transport
33/334 monoatomic anion transport
4/27 regulation of membrane lipid distribution
1/12 regulation of sterol transport
25/236 monoatomic anion transmembrane transport
10/110 xenobiotic transport
2/29 acid secretion
7/47 vascular process in circulatory system
16/112 circulatory system process
1/10 vascular transport
4/14 body fluid secretion
29/199 regulation of transmembrane transport
1/17 obsolete regulation of anion channel
2/13 regulation of chloride transport
21/212 epithelial cell differentiation
19/147 epithelial cell development
2/10 establishment of endothelial barrier
6/38 digestion
6/49 monoatomic anion homeostasis
5/15 intracellular monoatomic anion homeostasis
3/85 lipid homeostasis
4/12 membrane protein proteolysis
10/93 tissue homeostasis
74/650 protein phosphorylation
3/26 obsolete activation of MAPKK
7/74 stress-activated protein kinase signaling cascade
59/212 translational initiation
3/10 translation reinitiation
86/275 cytoplasmic translation
7/21 formation of translation preinitiation complex
111/684 amide biosynthetic process
18/159 protein-containing complex disassembly
80/607 ribonucleoprotein complex biogenesis
4/36 ribosomal small subunit assembly
55/416 ribosome biogenesis
20/117 ribosome assembly
50/370 protein-RNA complex organization
44/332 rRNA metabolic process
14/112 maturation of SSU-rRNA
59/591 obsolete ncRNA metabolic process
44/441 obsolete ncRNA processing
18/140 ribosomal small subunit biogenesis
20/125 ribosomal large subunit biogenesis
52/189 protein targeting to membrane
77/490 establishment of protein localization to organelle
54/226 establishment of protein localization to membrane
67/417 localization within membrane
74/374 nucleic acid catabolic process
54/204 nuclear-transcribed mRNA catabolic process, nonsense-mediated decay
100/599 obsolete organic cyclic compound catabolic process
3/29 cellular response to interleukin-4
9/61 nuclear receptor-mediated signaling pathway
4/15 regulation of retinoic acid receptor signaling pathway
8/49 steroid hormone receptor signaling pathway
7/17 response to testosterone
4/16 regulation of substrate adhesion-dependent cell spreading
54/422 regulation of amide metabolic process
4/18 ATF6-mediated unfolded protein response
42/305 protein folding
24/140 obsolete chaperone-mediated protein folding
11/43 ER-nucleus signaling pathway
34/297 nuclear transport
26/205 nuclear export
21/123 mRNA cis splicing, via spliceosome
59/498 mRNA processing
4/19 cytoplasm protein quality control
8/57 response to misfolded protein
13/111 obsolete regulation of cysteine-type endopeptidase
40/422 regulation of proteolysis
14/131 regulation of peptidase
9/45 obsolete regulation of telomerase
9/23 proteasomal ubiquitin-independent protein catabolic process
4/22 fat cell differentiation

p < 1e-04
p < 0.001
p < 0.01

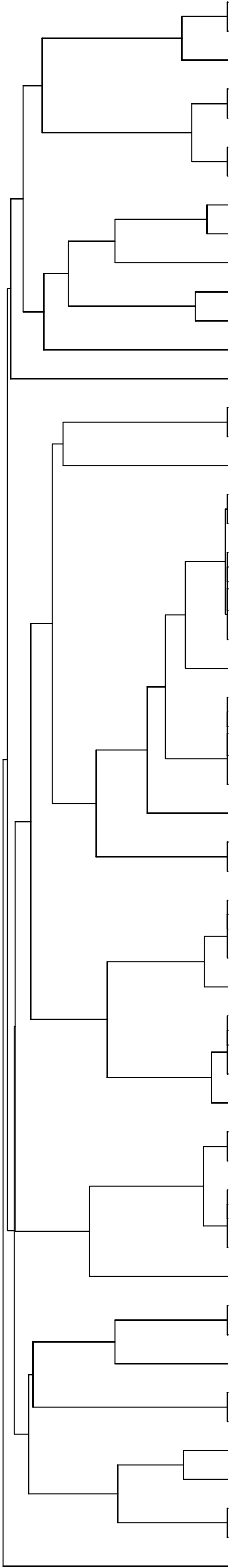
SYMBIONT WINTER – Biological Process (BP)



- 4/22 neuron projection membrane
- 35/353 dendritic tree
- 4/16 dendrite membrane
- 8/50 leading edge membrane
- 48/332 obsolete intrinsic component of plasma membrane
- 5/35 voltage-gated potassium channel complex
- 3/30 calcium channel complex
- 17/105 monoatomic ion channel complex
- 21/171 plasma membrane protein complex
- 6/20 brush border membrane
- 15/103 cell projection membrane
- 38/341 plasma membrane region
- 27/189 apical part of cell
- 18/120 apical plasma membrane
- 3/11 intercellular canaliculus
- 8/70 sarcolemma
- 7/51 T-tubule
- 15/90 I band
- 2/31 extrinsic component of organelle membrane
- 9/128 extrinsic component of membrane
- 42/485 endosome
- 58/298 external encapsulating structure
- 2/22 spore wall
- 2/67 ascospore-type prospore
- 3/23 amyloplast
- 8/99 cell cortex region
- 29/318 cell cortex
- 1/17 new growing cell tip
- 64/343 plasmodesma
- 75/459 cell-cell junction
- 7/26 actin filament bundle
- 18/104 apoplast
- 1/28 phragmoplast
- 5/25 endoplasmic reticulum-Golgi intermediate compartment membrane
- 15/96 endoplasmic reticulum lumen
- 46/125 obsolete polysomal ribosome
- 57/205 obsolete polysome
- 48/143 large ribosomal subunit
- 77/303 ribosome
- 76/251 cytosolic ribosome
- 23/84 small ribosomal subunit
- 16/60 rough endoplasmic reticulum membrane
- 17/86 rough endoplasmic reticulum
- 22/98 preribosome, large subunit precursor
- 35/196 preribosome
- 10/32 proteasome storage granule
- 14/49 cytosolic proteasome complex
- 97/345 obsolete cytosolic part
- 9/23 proteasome core complex
- 24/112 peptidase complex
- 6/14 proteasome core complex, alpha-subunit complex
- 77/613 intracellular protein-containing complex
- 4/16 cytoplasmic ubiquitin ligase complex
- 6/12 multi-eIF complex
- 5/44 azurophil granule lumen
- 8/62 azurophil granule
- 10/39 respiratory chain complex
- 8/27 obsolete respirasome
- 14/52 inner mitochondrial membrane protein complex
- 21/126 mitochondrial protein-containing complex
- 16/113 Sm-like protein family complex
- 4/31 U1 snRNP
- 10/45 U2 snRNP
- 22/144 U2-type spliceosomal complex
- 33/271 spliceosomal complex
- 9/58 prespliceosome
- 9/64 precatalytic spliceosome
- 20/162 catalytic step 2 spliceosome
- 8/53 U2-type catalytic step 2 spliceosome
- 4/45 U4/U6 x U5 tri-snRNP complex
- 60/565 ribonucleoprotein granule
- 30/258 nuclear ribonucleoprotein granule
- 6/24 catalytic step 1 spliceosome
- 4/39 sno(s)RNA-containing ribonucleoprotein complex
- 7/10 enzyme activator complex
- 3/14 new mitotic spindle pole body

p < 1e-04
p < 0.001
p < 0.01

SYMBIONT WINTER – Cellular Component (CC)



1/14 ligase regulator

32/248 calcium ion binding

4/27 structural constituent of muscle

11/62 rRNA binding

2/22 5S rRNA binding

78/339 structural molecule

63/200 structural constituent of ribosome

92/507 mRNA binding

4/11 opsonin binding

27/230 ubiquitin-like protein ligase binding

85/574 protein-containing complex binding

3/14 protein kinase C inhibitor

12/57 translation initiation factor

19/139 GTPase

12/71 obsolete inorganic anion transmembrane transporter

22/216 monoatomic anion transmembrane transporter

0/28 ammonium channel

20/146 gated channel

12/110 voltage-gated channel

6/58 potassium channel

28/184 monoatomic ion channel

67/540 monoatomic ion transmembrane transporter

29/216 channel

9/51 ligand-gated monoatomic ion channel

47/392 monoatomic cation transmembrane transporter

4/49 calcium channel

13/108 calcium ion transmembrane transporter

29/241 metal ion transmembrane transporter

11/106 potassium ion transmembrane transporter

16/89 sodium ion transmembrane transporter

4/16 P-type sodium transporter

3/39 phospholipid transporter

7/67 lipid transporter

1/15 phosphatidylcholine transporter

2/13 lipase

46/337 active transmembrane transporter

34/201 primary active transmembrane transporter

6/40 ABC-type transporter

3/14 xenobiotic transmembrane transporter

69/620 ATP-dependent

14/111 ATP-dependent, acting on DNA

13/230 nucleic acid conformation isomerase

5/63 DNA helicase

32/366 macromolecular conformation isomerase

19/142 polypeptide conformation or assembly isomerase

9/35 cyclic nucleotide-dependent protein kinase

65/627 phosphotransferase, alcohol group as acceptor

3/18 MAP kinase kinase kinase

16/110 molecular transducer

12/64 transmembrane signaling receptor

10/46 cyclic nucleotide binding

77/648 adenylyl ribonucleotide binding

3/17 3',5'-cyclic-AMP phosphodiesterase

4/23 cyclic-nucleotide phosphodiesterase

4/14 phosphotransferase, carboxyl group as acceptor

p < 1e-04

p < 0.001

p < 0.01

SYMBIONT WINTER – Molecular Function (MF)